JEE Mains 2019 Chapter wise Question Bank

Statistics – Questions

Q1

5 students of a class have an average height 150 cm and variance 18 cm^2. A new student, whose height is 156 cm, joined them. The variance (in cm^2) of the height of these six students is:

(1) 16 (2) 22
(3) 20 (4) 18

9 Jan Morning

Q2

A data consists of n observations:

$x_1, x_2, ..., x_n$. If $\sum_{i=1}^{n}(x_i+1)^2 = 9n$ and $\sum_{i=1}^{n}(x_i-1)^2 = 5n$

then the standard deviation of this data is:

(1) 2 (2) $\sqrt{5}$
(3) 5 (4) $\sqrt{7}$

9 Jan Evening

Q3

The mean of five observations is 5 and their variance is 9.20. If three of the given five observations are 1, 3 and 8, then a ratio of other two observations is:

(1) 10 : 3 (2) 4 : 9
(3) 5 : 8 (4) 6 : 7

10 Jan Morning

Q4

If mean and standard deviation of 5 observations x_1, x_2, x_3, x_4, x_5 are 10 and 3, respectively, then the variance of 6 observations $x_1, x_2, ..., x_5$ and – 50 is equal to:

(1) 509.5 (2) 586.5
(3) 582.5 (4) 507.5

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Q5

The outcome of each of 30 items was observed; 10 items gave an outcome $\frac{1}{2}-d$ each, 10 items gave outcome $\frac{1}{2}$ each and the remaining 10 items gave outcome $\frac{1}{2}+d$ each.

If the variance of this outcome data is $\frac{4}{3}$ then $|d|$ equals :

(1) $\frac{2}{3}$ (2) 2
(3) $\frac{\sqrt{5}}{2}$ (4) $\sqrt{2}$

11 Jan Morning

Q6

A bag contains 30 white balls and 10 red balls. 16 balls are drawn one by one randomly from the bag with replacement. If X be the number of white balls drawn, then

$\left(\frac{\text{mean of X}}{\text{standard deviation of X}}\right)$ is equal to:

(1) 4 (2) $4\sqrt{3}$ (3) $3\sqrt{2}$ (4) $\frac{4\sqrt{3}}{3}$

11 Jan Evening

Q7

If the sum of the deviations of 50 observations from 30 is 50, then the mean of these observations is :

(1) 30 (2) 51
(3) 50 (4) 31

12 Jan Morning

Q8

There are m men and two women participating in a chess tournament. Each participant plays two games with every other participant. If the number of games played by the men between themselves exceeds the number of games played between the men and the women by 84, then the value of m is

(1) 12 (2) 11

(2) 9 (4) 7

12 Jan Evening

Q9

The mean and the variance of five observations are 4 and 5.20, respectively. If three of the observations are 3, 4 and 4; then the absolute value of the difference of the other two observations, is :

(1) 7 (2) 5

(3) 1 (4) 3

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Q10

The mean and variance of seven observations are 8 and 16, respectively. If 5 of the observations are 2, 4, 10, 12, 14, then the product of the remaining two observations is :

(1) 45 (2) 49 (3) 48 (4) 40

8 April Morning

Q11

A student scores the following marks in five tests: 45, 54, 41, 57, 43. His score is not known for the sixth test. If the mean score is 48 in the six tests, then the standard deviation of the marks in six tests is :

(1) $\frac{10}{\sqrt{3}}$ (2) $\frac{100}{3}$ (3) $\frac{10}{3}$ (4) $\frac{100}{\sqrt{3}}$

8 April Evening

Q12

If the standard deviation of the numbers –1, 0, 1, k is $\sqrt{5}$ where k > 0, then k is equal to:

(1) $2\sqrt{6}$ (2) $2\sqrt{\frac{10}{3}}$

(3) $4\sqrt{\frac{5}{3}}$ (4) $\sqrt{6}$

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Q13

The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to:

(1) 9/4 (2) 7/2 (3) 8/3 (4) 7/3

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Q14

If for some x $\in$ **R**, the frequency distribution of the marks obtained by 20 students in a test is :

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x-5$	x^2-3x	x

then the mean of the marks is :

(1) 3.2 (2) 3.0 (3) 2.5 (4) 2.8

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Q15

If both the mean and the standard deviation of 50 observations $x_1, x_2, \ldots, x_{50}$ are equal to 16, then the mean of $(x_1-4)^2, (x_2-4)^2, \ldots, (x_{50}-4)^2$ is :

(1) 400 (2) 380 (3) 525 (4) 480

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Q16

If the data $x_1, x_2, \ldots, x_{10}$ is such that the mean of first four of these is 11, the mean of the remaining six is 16 and the sum of squares of all of these is 2,000 ; then the standard deviation of this data is :

(1) $2\sqrt{2}$ (2) 2 (3) 4 (4) $\sqrt{2}$

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Statistics - Solutions

Q1

(3) $\because$ Variance $= \sigma^2 = \dfrac{\Sigma x_i^2}{N} - (\overline{x})^2$

$\Rightarrow \quad 18 = \dfrac{\Sigma x_i^2}{5} - (150)^2$

$\Rightarrow \quad \Sigma x_i^2 = 90 + 112590 = 112590$

Then, variance of the height of six students

$$V' = \frac{112590 + (156)^2}{6} - \left(\frac{750+156}{6}\right)^2$$

$= 22821 - 22801$

$= 20$

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Q2

(2) Variance is given by,

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)^2$$

$$\sigma^2 = \frac{1}{n}A - \frac{1}{n^2}B^2 \quad \text{...(i)}$$

Here, $A = \sum_{i=1}^{n} x_i^2$ and $B = \sum_{i=1}^{n} x_i$

$\because \quad \sum_{i=1}^{n}(x_i+1)^2 = 9n$

$\Rightarrow \quad A + n + 2B = 9n \Rightarrow A + 2B = 8n$ ...(ii)

$\because \quad \sum_{i=1}^{n}(x_i-1)^2 = 5n$

$\Rightarrow \quad A + n - 2B = 5n \Rightarrow A - 2B = 4n$ ...(iii)

From (ii) and (iii),

$A = 6n, B = n$

$\Rightarrow \sigma^2 = \dfrac{1}{n} \times 6n - \dfrac{1}{n^2} \times n^2 = 6 - 1 = 5$

$\Rightarrow \sigma = \sqrt{5}$

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Q3

(2) Since mean of x_1, x_2, x_3, x_4 and x_5 is 5

$\Rightarrow x_1 + x_2 + x_3 + x_4 + x_5 = 25$

$\Rightarrow 1 + 3 + 8 + x_4 + x_5 = 25$

$\Rightarrow x_4 + x_5 = 13$ ...(1)

$$\because \frac{\sum_{i=1}^{5} x_i^2}{5} - (5)^2 = 9.2 \Rightarrow \sum_{i=1}^{5} x_i^2 = 5(25 + 9.2)$$

$= 125 + 46 = 171$

$\Rightarrow (1)^2 + (3)^2 + (8)^2 + x_4^2 + x_5^2 = 171$

$\Rightarrow x_4^2 + x_5^2 = 97$ ...(2)

$\Rightarrow (x_4 + x_5)^2 - 2x_4 x_5 = 97$

$\Rightarrow 2x_4 x_5 = 13^2 - 97 = 72 \Rightarrow x_4 x_5 = 36$ ...(3)

(1) and (3) $\Rightarrow x_4 : x_5 = \frac{4}{9}$ or $\frac{9}{4}$

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Q4

$$\textbf{(4)} \because \bar{x} = \frac{\sum_{i=1}^{5} x_i}{5} \Rightarrow \sum_{i=1}^{5} x_i = 10 \times 5 = 50 \Rightarrow \sum_{i=1}^{6} xi = 50 - 50 = 0$$

$$\frac{\sum_{i=1}^{5} x_i^2}{5} - (10)^2 = 3^2 = 9$$

$$\Rightarrow \sum_{i=1}^{5} x_i^2 = 545$$

Then,

$$\Rightarrow \sum_{i=1}^{6} x_i^2 = \sum_{i=1}^{5} x_i^2 + (-50)^2$$

$= 545 + (-50)^2 = 3045$

$$\text{Variance} = \frac{\sum_{i=1}^{6} x_i^2}{6} - \left(\frac{\sum_{i=1}^{6} x_i}{6}\right)^2 = \frac{3045}{6} - 0 = 507.5$$

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Q5

(4) Outcomes are $\left(\frac{1}{2} - d\right), \left(\frac{1}{2} - d\right), \ldots,$ 10 times, $\frac{1}{2}, \frac{1}{2}, \ldots,$ 10 times, $\frac{1}{2} + d, \frac{1}{2} + d, \ldots,$ 10 times

$$\text{Mean} = \frac{1}{30}\left(\frac{1}{2} \times 30\right) = \frac{1}{2}$$

Variance of the outcomes is,

$$\sigma^2 = \frac{1}{30}\Sigma x_i^2 - (\bar{x})$$

$$= \frac{1}{30}\left[\left(\frac{1}{2} - d\right)^2 \times 10 + \left(\frac{1}{2}\right)^2 \times 10 + \left(\frac{1}{2} + d\right)^2 \times 10\right] - \frac{1}{4}$$

$$\Rightarrow \frac{4}{3} = \frac{1}{30}\left[30 \times \frac{1}{4} + 20d^2\right] - \frac{1}{4}$$

$$\Rightarrow \frac{4}{3} = \frac{1}{4} + \frac{2}{3}d^2 - \frac{1}{4}$$

$$\Rightarrow d^2 = 2 \Rightarrow |d| = \sqrt{2}$$

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Q6

(2) $P(\text{white ball}) = \frac{30}{40} = \frac{3}{4}$, $Q(\text{red ball}) = \frac{10}{40} = \frac{1}{4}$, $n = 16$

$$\frac{\text{Mean of } X}{\text{standard deviation of } X} = \frac{nP}{\sqrt{nPQ}} = \frac{\sqrt{nP}}{\sqrt{Q}}$$

$$= \sqrt{\frac{16 \times \frac{3}{4}}{\frac{1}{4}}} = \sqrt{48} = 4\sqrt{3}$$

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Q7

(4) Given, $\sum_{i=1}^{50}(x_i - 30) = 50$

$$\sum_{i=1}^{50} x_i - 50(30) = 50$$

$$\Rightarrow \quad \sum_{i=1}^{50} x_i = 1550$$

Mean, $\bar{x} = \frac{\sum x_i}{50}$

$$= \frac{1550}{50}$$

$$= 31$$

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Q8

(1) ${}^mC_2 \times 2 = {}^mC_1 \cdot {}^2C_1 \times 2 + 84$

$$m(m-1) = 4m + 84$$

$$m^2 - 5\text{m} - 84 = 0$$

$$m^2 - 12m - 7m - 84 = 0$$

$$m(m-12) + 7(m-12) = 0$$

$$m = 12, m = -7$$

$\because \quad m > 0$

$$m = 12$$

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Q9

(1) Let two observations be x_1 and x_2, then

$$\frac{x_1 + x_2 + 3 + 4 + 4}{5} = 4$$

$$\Rightarrow \quad x_1 + x_2 = 9 \quad \text{...(1)}$$

$$\text{Variance} = \frac{\sum x_i^2}{N} - (\bar{x})^2$$

$$(5 \cdot 20) = \frac{9 + 16 + 16 + x_1^2 + x_2^2}{5} - 16$$

$$26 = 41 + x_1^2 + x_2^2 - 80$$

$$x_1^2 + x_2^2 = 65 \quad \text{...(2)}$$

From (1) and (2);

$$x_1 = 8, x_2 = 1$$

Hence, the required value of the difference of other two observations $= |x_1 - x_2| = 7$

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Q10

(3) Let the remaining numbers are a and b.

$$\text{Mean } (\bar{x}) = \frac{\sum x_i}{N} = \frac{2 + 4 + 10 + 12 + 14 + a + b}{7} = 8$$

$$\Rightarrow a + b = 14 \quad \text{...(i)}$$

$$\text{Variance } (\sigma^2) = \frac{\sum x_i^2}{N} - (\bar{x})^2 = 16$$

$$\Rightarrow \frac{2^2 + 4^2 + 10^2 + 12^2 + 14^2 + a^2 + b^2}{7} - (8)^2 = 16$$

$$\Rightarrow a^2 + b^2 = 100 \quad \text{...(ii)}$$

From (i) and (ii), $(14 - b)^2 + b^2 = 100$

$$\Rightarrow 196 + \text{b}^2 - 28b + b^2 = 100$$

$$\Rightarrow b^2 - 14b + 48 = 0$$

$$\Rightarrow b = 6, 8$$

$\therefore \quad a = 8, 6.$

$\therefore \quad (a, b) = (6, 8)$ or $(8, 6)$

Hence, the product of the remaining two observations $= ab = 48$

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Q11

(1) $\because$ Mean score = 48

Let unknown score be x,

$$\therefore\ \bar{x} = \frac{41+45+54+57+43+x}{6} = 48$$

$$\Rightarrow\ x + 240 = 288 \quad \Rightarrow x = 48$$

$$\text{Now, } \sigma^2 = \frac{1}{6}[(48-41)^2 + (48-45)^2 + (48-54)^2 + (48-57)^2 + (48-43)^2 + (48-48)^2$$

$$= \frac{1}{6}(49+9+36+81+25) = \frac{200}{6} = \frac{100}{3}$$

$$\sigma = \frac{10}{\sqrt{3}}$$

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Q12

(1) Mean of given observation $= \frac{k}{4}$

$\because$ Standard deviation = 5

$\therefore\ \sigma^2 = 5$ ...(1)

$$\Rightarrow\ \sigma^2 = \frac{1}{n}\Sigma(x_i - \bar{x})^2 = \frac{\left(\frac{k}{4}+1\right)^2 + \left(\frac{k}{4}\right)^2 + \left(\frac{k}{4}-1\right)^2 + \left(\frac{3k}{4}\right)^2}{4} = 5$$

...(2)

$$\Rightarrow\ \frac{\frac{12k^2}{16}+2}{4} = 5$$

$$\Rightarrow k = 2\sqrt{6}$$

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Q13

(4) Ten numbers in increasing order are

10, 22, 26, 29, 34, x, 42, 67, 70, y

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{x+y+300}{10} = 42 \ \Rightarrow x + y = 120$$

$$\text{Median} = \frac{T_5 + T_6}{2} = 35 = \frac{34+x}{2} \ \Rightarrow x = 36 \text{ and } y = 84$$

$$\text{Hence, } \frac{y}{x} = \frac{84}{36} = \frac{7}{3}$$

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Q14

(4) Number of students are,

$(x+1)^2 + (2x-5) + (x^2 - 3x) + x = 20$

$\Rightarrow 2x^2 + 2x - 4 = 20 \Rightarrow x^2 + x - 12 = 0$

$\Rightarrow (x+4)(x-3) = 0 \Rightarrow x = 3$

$\therefore$

Marks	2	3	5	7
No. of students	16	1	0	3

$$\text{Average marks} = \frac{32+3+21}{20} = \frac{56}{20} = 2.8$$

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Q15

(1) Given, mean and standard deviation are equal to 16.

$$\therefore \frac{x_1 + x_2 + \ldots. x_{50}}{50} = 16$$

$$\text{and } 16^2 = \frac{x_1^2 + x_2^2 \ldots x_{50}^2}{50} - 16^2$$

$$\Rightarrow 2\,(16)^2\,50 = x_1^2 + x_2^2 + \ldots. x_{50}^2$$

$$\text{Required mean} = \frac{(x_1-4)^2 + (x_2-4)^2 + \ldots(x_{50}-4)^2}{50}$$

$$= \frac{x_1^2 + x_2^2 + \ldots + x_{50}^2 + 50\times16 - 8(x_1 + x_2 + \ldots + x_{50})}{50}$$

$$= \frac{16^2(100) + (50) - 8(16\times50)}{50} = 400$$

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Q16

(2) According to the question,

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = 11 \Rightarrow x_1 + x_2 + x_3 + x_4 = 44$$

$$\frac{x_5 + x_6 + \ldots + x_{10}}{6} = 16 \Rightarrow x_5 + x_6 + \ldots + x_{10} = 96$$

$$\text{and } x_1^2 + x_2^2 + \ldots x_{10}^2 = 2000$$

$$\because \text{ standard deviation, } \sigma^2 = \frac{\sum x_i^2}{N} - (\bar{x})^2$$

$$= \frac{2000}{10} - \left(\frac{44+96}{10}\right)^2 = 4 \ \Rightarrow \sigma = 2$$

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